

XCS234 Assignment 4

Due Sunday, October 12 at 11:59pm PT.

Guidelines

1. If you have a question about this homework, we encourage you to post your question on our Slack channel, at <http://xcs234-scpd.slack.com/>
2. Familiarize yourself with the collaboration and honor code policy before starting work.
3. For the coding problems, you must use the packages specified in the provided environment description. Since the autograder uses this environment, we will not be able to grade any submissions which import unexpected libraries.

Submission Instructions

Written Submission: Some questions in this assignment require a written response. For these questions, you should submit a PDF with your solutions online in the online student portal. As long as the PDF is legible and organized, the course staff has no preference between a handwritten and a typeset L^AT_EX submission. If you wish to typeset your submission and are new to L^AT_EX, you can get started with the following:

- Type responses only in `submission.tex`.
- Submit the compiled PDF, **not** `submission.tex`.
- Use the commented instructions within the `Makefile` and `README.md` to get started.

Coding Submission: Some questions in this assignment require a coding response. For these questions, you should submit only the `src/submission.py` file in the online student portal. For further details, see Writing Code and Running the Autograder below.

Honor code

We strongly encourage students to form study groups. Students may discuss and work on homework problems in groups. However, each student must write down the solutions independently, and without referring to written notes from the joint session. In other words, each student must understand the solution well enough in order to reconstruct it by him/herself. In addition, each student should write on the problem set the set of people with whom s/he collaborated. Further, because we occasionally reuse problem set questions from previous years, we expect students not to copy, refer to, or look at the solutions in preparing their answers. It is an honor code violation to intentionally refer to a previous year's solutions. For SCPD classes, it is also important that students avoid opening pull requests containing their solution code on the shared assignment repositories. More information regarding the Stanford honor code can be found at <https://communitystandards.stanford.edu/policies-and-guidance/honor-code>.

Writing Code and Running the Autograder

All your code should be entered into `src/submission.py`. When editing these files, please only make changes between the lines containing `### START_CODE_HERE ###` and `### END_CODE_HERE ###`. Do NOT make changes to files other than `src/submission.py`.

The unit tests in `src/grader.py` (the autograder) will be used to verify a correct submission. Run the autograder locally using the following terminal command within the `src/` subdirectory:

```
$ python grader.py
```

There are two types of unit tests used by the autograder:

- **basic:** These tests are provided to make sure that your inputs and outputs are on the right track, and can be run locally.

- **hidden:** These unit tests are NOT visible locally. These hidden tests will be run when you submit your code to the Gradescope autograder via the online student portal, and will provide feedback on how many points you have earned. These tests will evaluate elements of the assignment, and run your code with more complex inputs and corner cases. Just because your code passed the basic local tests does not necessarily mean that they will pass all of the hidden tests.

For debugging purposes, you can run a single unit test locally. For example, you can run the test case `3a-0-basic` using the following terminal command within the `src/` subdirectory:

```
$ python grader.py 3a-0-basic
```

Before beginning this course, please walk through the [Anaconda Setup for XCS Courses](#) to familiarize yourself with the coding environment. Use the env defined in `src/environment.yml` to run your code. This is the same environment used by the online autograder.

Test Cases

The autograder is a thin wrapper over the python `unittest` framework. It can be run either locally (on your computer) or remotely (on SCPD servers). The following description demonstrates what test results will look like for both local and remote execution. For the sake of example, we will consider two generic tests: `1a-0-basic` and `1a-1-hidden`.

Local Execution - Basic Tests

When a basic test like `1a-0-basic` passes locally, the autograder will indicate success:

```
----- START 1a-0-basic: Basic test case.
----- END 1a-0-basic [took 0:00:00.000062 (max allowed 1 seconds), 2/2 points]
```

When a basic test like `1a-0-basic` fails locally, the error is printed to the terminal, along with a stack trace indicating where the error occurred:

```
----- START 1a-0-basic: Basic test case.
<class 'AssertionError'>
{'a': 2, 'b': 1} != None ← This error caused the test to fail.
File "/Users/grinch/Local_Documents/Software/anaconda3/envs/XCS221/lib/python3.6/unittest/case.py", line 59, in testPartExecutor
    yield
File "/Users/grinch/Local_Documents/Software/anaconda3/envs/XCS221/lib/python3.6/unittest/case.py", line 605, in run
    testMethod()
File "/Users/grinch/Local_Documents/SCPD/XCS221/A1/src/graderUtil.py", line 54, in wrapper
    result = func(*args, **kwargs)
File "/Users/grinch/Local_Documents/SCPD/XCS221/A1/src/graderUtil.py", line 83, in wrapper
    result = func(*args, **kwargs)
File "/Users/grinch/Local_Documents/SCPD/XCS221/A1/src/grader.py", line 23, in test_0
    submission.extractWordFeatures("a b a") ← In this case, start your debugging
                                           in line 23 of grader.py.
File "/Users/grinch/Local_Documents/Software/anaconda3/envs/XCS221/lib/python3.6/unittest/case.py", line 829, in assertEquals
    assertion_func(first, second, msg=msg)
File "/Users/grinch/Local_Documents/Software/anaconda3/envs/XCS221/lib/python3.6/unittest/case.py", line 822, in _baseAssertEqual
    raise self.failureException(msg)
----- END 1a-0-basic [took 0:00:00.003809 (max allowed 1 seconds), 0/2 points]
```

Remote Execution

Basic and hidden tests are treated the same by the remote autograder, however the output of hidden tests will only appear once you upload your code to GradeScope. Here are screenshots of failed basic and hidden tests. Notice that the same information (error and stack trace) is provided as the in local autograder, now for both basic and hidden tests.

1a-0-basic) Basic test case. (0.0/2.0)

```
<class 'AssertionError': {'a': 2, 'b': 1} != None
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 59, in testPartExecutor
    yield
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 605, in run
    testMethod()
File "/autograder/source/graderUtil.py", line 54, in wrapper
    result = func(*args, **kwargs)
File "/autograder/source/graderUtil.py", line 83, in wrapper
    result = func(*args, **kwargs)
File "/autograder/source/grader.py", line 23, in test_0
    submission.extractWordFeatures("a b a"))
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 829, in assertEqual
    assertion_func(first, second, msg=msg)
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 822, in _baseAssertEqual
    raise self.failureException(msg)
```

Just like in the local autograder, this error caused the test to fail.

Just like in the local autograder, start your debugging in line 23 of grader.py.

1a-1-hidden) Test multiple instances of the same word in a sentence. (0.0/3.0)

```
<class 'AssertionError': {'a': 23, 'ab': 22, 'aa': 24, 'c': 16, 'b': 15} != None
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 59, in testPartExecutor
    yield
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 605, in run
    testMethod()
File "/autograder/source/graderUtil.py", line 54, in wrapper
    result = func(*args, **kwargs)
File "/autograder/source/graderUtil.py", line 83, in wrapper
    result = func(*args, **kwargs)
File "/autograder/source/grader.py", line 31, in test_1
    self.compare_with_solution_or_wait(submission, 'extractWordFeatures', lambda f: f(sentence))
File "/autograder/source/graderUtil.py", line 183, in compare_with_solution_or_wait
    self.assertEqual(ans1, ans2)
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 829, in assertEqual
    assertion_func(first, second, msg=msg)
File "/autograder/source/miniconda/envs/XCS221/lib/python3.6/unittest/case.py", line 822, in _baseAssertEqual
    raise self.failureException(msg)
```

This error caused the test to fail.

Start your debugging in line 31 of grader.py.

Finally, here is what it looks like when basic and hidden tests pass in the remote autograder.

1a-0-basic) Basic test case. (2.0/2.0)

1a-1-hidden) Test multiple instances of the same word in a sentence. (3.0/3.0)

0 Introduction

In this assignment, we will be using the [Hopper environment](#) where we will first derive reward functions for specific tasks, moving then to adding human preferences to learn the reward function and then ending the process with an end-to-end approach, where we skip the reward learning and reinforcement learning steps, and optimize the model directly from preference data.

Deliverables:

For this assignment, please submit the following files to gradescope to receive points for coding questions:

- `src/submission/__init__.py`
- `src/submission/rlhf.py`
- `src/submission/dpo.py`

1 Reward engineering

In Assignment 3 you applied Policy Gradient methods to solve an environment with a provided reward function. The process of deriving a reward function for a specific task is called reward engineering.

(a) [1 point (Written)]

Why is reward engineering usually hard? What are potential risks that come with specifying an incorrect reward function? Provide an example of a problem and a reward function that appears to be adequate but may have unintended consequences.

It's generally very difficult to translate high level objectives into a math mathematical reward function an agent can optimize and it is very difficult to avoid reward hacking that results in an unintended consequences.

Example: Content recommendation agent

The reward function could be number of clicks. The unintended result could be recommending all the clickbait contents instead of high quality, actually interesting contents.

(b) [1 point (Written)]

Read the description of the [Hopper environment](#). Using your own words, describe the goal of the environment, and how each term of the reward functions contributes to encourage the agent to achieve it. (**Optional:** Do you agree with this reward function? Would you change it? How?)

It's a single legged robot trying to hop forward.

$$\text{reward} = \text{healthy_reward} + \text{forward_reward} - \text{ctrl_cost}$$

Where

healthy_reward : Keep the agent to stand by keeping the torso in a healthy state

forward_reward : Push the agent to move forward by maximize the x-velocity

ctrl_cost : Prevent the agent to aggressively move, i.e. move smoothly by penalize the large control

(c) [1 point (Written)]

By default, the episode terminates when the agent leaves the set of “healthy” states. What do these “healthy” states mean? Name one advantage and one disadvantage of this early termination.

It means the hopper is standing and not falling over. It is determine by two factors: the torso's height must be above 0.8 meters, and its angle must be within 0.2 radians of vertical.

Advantage of early-termination: speed up training.

Disadvantage of early-termination: can't learn to recover from falling.

(d) [0 points (Coding)]

Use the provided starter code to train a policy using PPO to solve the Hopper environment for 3 different seeds. Do this with and without early termination.

```
$ python ppo_hopper.py [--early-termination] --seed SEED
```

(e) [1 point (Written)]

The plot of the episodic returns along training, with and without early termination is seen below. You can generate the plot by running

```
$ python plot.py --directory results --output ppo_hopper.png --seeds SEEDS
```

where SEEDS is a comma-separated list of the seeds you used. Comment on the performance in terms of training epochs and wall time. Is the standard error in the average returns high or low? How could you obtain a better estimate of the average return on Hopper achieved by a policy optimized with PPO?

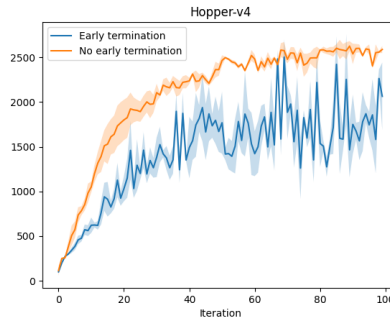
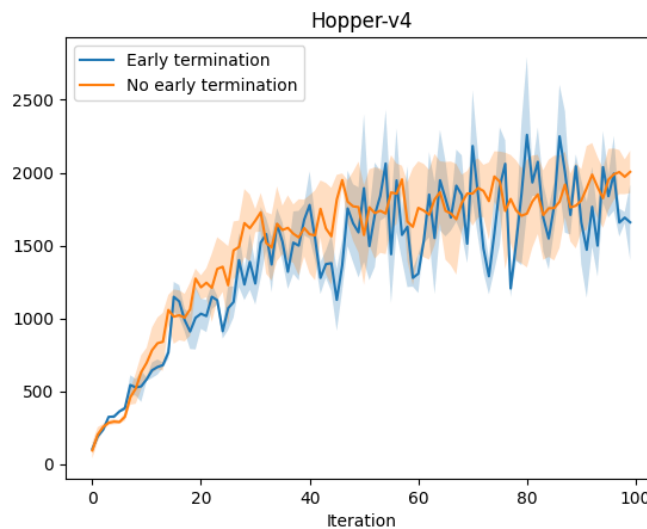


Figure 1: PPO results



The performance without early-termination is better than with early-termination. It also more consistent, i.e. the standard error is smaller. To get better estimate, we should sample more times by using different random seeds (at least 10-20 times).

(f) [1 point (Written)]

Pick one of the trained policies and render a video of an evaluation rollout.

```
$ python render.py --checkpoint results/[MODEL]/model.zip
```

Does the agent successfully complete the assigned task? Does it complete it in the way you would expect it to, or are you surprised by the agent behavior?

I looked at a rendered video of the agent trained without early-termination.

The agent seem to learn some basic control an able to move forward. But it also keep repeatedly falling and trying to stand up again.

It partially complete the task because it's moving. But it fails to keep the agent in a healthy state. It doesn't seem to be too surprised because the reward function doesn't care about the continuous time it keep in the healthy state.

(g) [1 point (Written)]

Render another video for another policy. How do the two rollouts compare? Do you prefer one over the other?

I looked at another without-early-termination model.

The agent quickly falling over with very little progress. It still able to crawling forward on the ground but never stand up again. I would consider this a completely failing.

2 Learning from Preferences

In the previous part you trained multiple policies from scratch and compared them at the end of training. In this section, we will see how we can use human preferences on two roll-outs to learn a reward function.

We will follow the framework proposed by [2]. A reward function $r : \mathcal{O} \times \mathcal{A} \rightarrow \mathbb{R}$ defines a preference relation $\succ$ if for all trajectories $\sigma^i = (o_t^i, a_t^i)_{t=0, \dots, T}$ we have that

$$((o_0^1, a_0^1), \dots, (o_T^1, a_T^1)) \succ ((o_0^2, a_0^2), \dots, (o_T^2, a_T^2))$$

whenever

$$r(o_0^1, a_0^1) + \dots + r(o_T^1, a_T^1) > r(o_0^2, a_0^2) + \dots + r(o_T^2, a_T^2).$$

Following the Bradley-Terry preference model [1], we can calculate the probability of one trajectory σ^1 being preferred over σ^2 as follows:

$$\hat{P}[\sigma^1 \succ \sigma^2] = \frac{\exp \sum \hat{r}(o_t^1, a_t^1)}{\exp \sum \hat{r}(o_t^1, a_t^1) + \exp \sum \hat{r}(o_t^2, a_t^2)},$$

where $\hat{r}$ is an estimate of the reward for a state-action pair. This is similar to a classification problem, and we can fit a function approximator to $\hat{r}$ by minimizing the cross-entropy loss between the values predicted with the above formula and ground truth human preference labels $\mu(1)$ and $\mu(2)$:

$$\text{loss}(\hat{r}) = - \sum_{(\sigma^1, \sigma^2, \mu) \in \mathcal{D}} \mu(1) \log \hat{P}[\sigma^1 \succ \sigma^2] + \mu(2) \log \hat{P}[\sigma^2 \succ \sigma^1].$$

Once we have learned the reward function¹, we can apply any policy optimization algorithm (such as PPO) to maximize the returns of a model under it.

(a) [4 points (Written)]

Let $\hat{r}(o, a) = \phi_w(o, a)$. Calculate $\nabla_w \text{loss}(\hat{r}(o, a))$.

Let $R_1 = \sum \hat{r}(o^1, a^1) = \sum \phi_w(o^1, a^1)$ and $R_2 = \sum \hat{r}(o^2, a^2) = \sum \phi_w(o^2, a^2)$.

We can express the Bradley-Terry in sigmoid form.

$$\begin{aligned} \hat{P}[\sigma^1 > \sigma^2] &= \frac{e^{R_1}}{e^{R_1} + e^{R_2}} \\ &= \frac{1}{1 + e^{-(R_1 - R_2)}} \\ &= \sigma(R_1 - R_2) \end{aligned}$$

Substitute into loss function

$$L = -(\mu(1) \log(\sigma(R_1 - R_2)) + \mu(2) \log(\sigma(R_2 - R_1)))$$

Let $z = R_1 - R_2$, applying chain rule

$$\begin{aligned} \nabla_w L &= [-\mu(1)(1 - \sigma(z)) + \mu(2)\sigma(z)] \nabla_w z \\ &= [-\mu(1) + \mu(1)\sigma(z) + \mu(2)\sigma(z)] \nabla_w z \\ &= [(\mu(1) + \mu(2))\sigma(z) - \mu(1)] \nabla_w z \\ &= [(\mu(1) + \mu(2))\sigma(R_1 - R_2) - \mu(1)] \nabla_w (R_1 - R_2) \\ &= [(\mu(1) + \mu(2))\sigma(\sum_t \phi_w(o_t^1, a_t^1) - \sum_t \phi_w(o_t^2, a_t^2)) - \mu(1)] (\sum_t \nabla_w \phi_w(o_t^1, a_t^1) - \sum_t \nabla_w \phi_w(o_t^2, a_t^2)) \end{aligned}$$

¹Recent work on RLHF for reinforcement learning suggests that the pairwise feedback provided by humans on partial trajectories may be more consistent with regret, and that the learned reward function may be better viewed as an advantage function. See Knox et al. AAAI 2024 "Learning optimal advantage from preferences and mistaking it for reward." <https://openreview.net/forum?id=euZXhbTmQ7>

(b) [2 points (Written)]

In this problem we are trying to solve the same task as in the previous part, but this time we will learn a reward function from a dataset of preferences rather than manually specifying a reward function.

Load one of the samples from the preference dataset we provide you, and render a video of the two trajectories using the following command

```
$ python render.py --dataset ./data/prefs-hopper.npz --idx IDX
```

where `IDX` is an index into the preference dataset (if omitted a sequence will be chosen at random). Bear in mind that each sequence in the dataset has 25 timesteps, which means that the resulting videos will have 0.2 seconds. Take note of which sequence was labeled as preferred (this information will appear in the name of the generated videos, but for the coming parts it is helpful to know that 0 means the first sequence was preferred, 1 means the second one, and 0.5 means neither is preferred over the other). Do you agree with the label (that is, if shown the two trajectories, would you have ranked them the same way they appear in the dataset, knowing that we are trying to solve the Hopper environment)?

I got the following pairs

3147 (equally preferred): agree, no obvious differences

(c) [2 points (Written)]

Repeat the previous question for 4 more samples, keeping track of whether you personally agree with the dataset preference label. Use this to estimate how much you agree with whoever ranked the trajectories. How much did you get? Based on this agreement estimate, would you trust a reward function learned on this data?

Four more pairs:

4191 (equally preferred): agree, no obvious differences

6599 (one is preferred): not agree, the non-preferred one has better ending ankle angle to land

6844 (one is preferred): similar to 6599

7324 (one is preferred): agree, the preferred one landed on its heel, which is a better position to initiate the next hop

I am not confident to say I trust this data because firstly the sample is very short, and I disagree some of the samples.

(d) [8 points (Coding)]

Implement the functions in the `RewardModel` class (`submission/rlhf.py`), which is responsible for learning a reward function from preference data.

Afterwards run the following command to train the reward model using the provided preference data:

```
$ python run_rlhf.py
```

(e) [3 points (Written)]

Train a model using PPO and the learned reward function with 3 different random seeds. The expected plots of the average returns for both the original reward function and the learned reward function are given below.

```
$ python plot.py --rlhf-directory results_rlhf --output results_rlhf/hopper_rlhf.png --seeds 0
```

Do the two correlate?

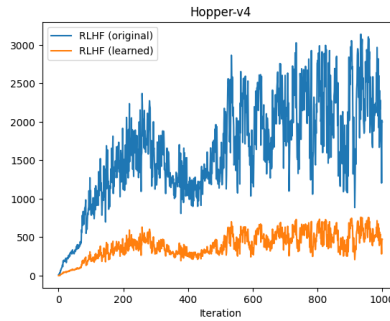
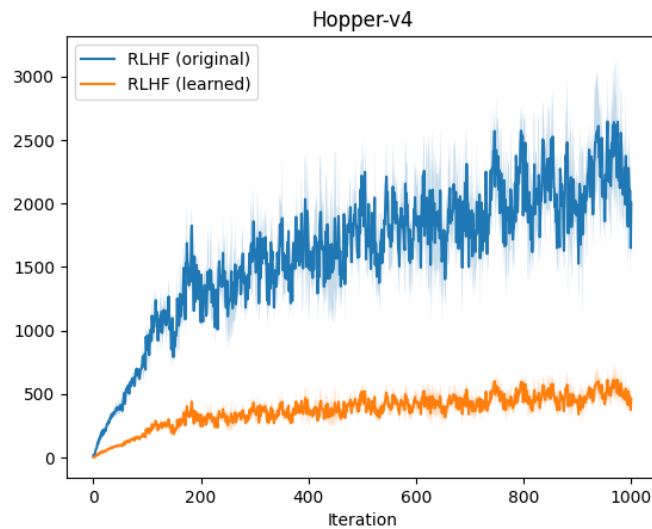


Figure 2: RLHF results



They are highly correlated.

(f) [2 points (Written)]

Is the learned reward function identifiable?

It does not identifiable because it has very similar trend as the original reward.

(g) [2 points (Written)]

Pick one of the policies and render a video of the agent behavior at the end of training.

```
$ python render.py --checkpoint ./results_rlhf/[MODEL]/model.zip
```

How does it compare to the behavior of the agent generated by policies trained from scratch? How does it compare to the demonstrations you've seen from the dataset?

It can perform more hops without falling. I think it is better in terms of hopping but the original policies can move forward more. The RLHF seems working as we prefer a landing position that is easier to initiate next hop.

3 Direct preference optimization (DPO)

In the previous question we saw how we could train a model based on preference data. However, suppose you are given a pre-trained model and the corresponding preference data. Ideally, you would like to optimize the model directly on the preference data, instead of having to learn a reward function, and then run PPO on it. That is the idea behind direct preference optimization (DPO) [3]. The algorithm proposed in the original paper allows us to skip the reward learning and reinforcement learning steps, and optimize the model directly from preference data by optimizing the following loss:

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right],$$

where π_{ref} is the policy from which we sampled y_w and y_l given x , π_{θ} is the policy we are optimizing, and σ is the sigmoid function.

We will use DPO in this question as well. To provide some context, let us consider the general approach for RLHF for text generation:

1. Train a large language model (LLM) to do next token prediction given a context (the tokens that came previously).
2. Given a fixed context x , generate possible next token sequence predictions y_1 and y_2 , and store the triple (x, y_1, y_2) .
3. Ask human supervisors to rank y_1 and y_2 given x according to individual preference.
4. Update the LLM to maximize the probability of giving the preferred answers using reinforcement learning.

In a similar way, given an observation x we could have two ranked sequences of actions $a_{1:T}^1$ and $a_{1:T}^2$, train the model to generate the preferred sequence of actions, and then execute them all². If the length of the generated action sequence is equal to the environment time horizon, this is called open-loop control. However, this approach lacks robustness, since the plan of actions will not change in response to disturbances or compounding errors. Instead, we are going to adapt this into a more robust scheme by training our policy to predict a sequence of actions for the next T time steps (where T is the length of the trajectories in our preference dataset), but only take the first action in the plan generated by the policy. In this way, we re-plan our actions at every time step, ensuring the ability to respond to disturbances.

(a) **[18 points (Coding)]**

Implement the `ActionSequenceModel` class instance methods. When called, the model should return a probability distribution for the actions over the number of next time steps specified at initialization. Use a multivariate normal distribution for each action, with mean and standard deviation predicted by a neural network (see the starter code for more details).³

(b) **[4 points (Coding)]**

Implement the `update` method of the `SFT` class. This class will be used to pre-train a policy on the preference data by maximizing the log probabilities of the preferred actions given the observations in the dataset.

(c) **[4 points (Coding)]**

Implement the `update` method of the `DPO` class. This should minimize the DPO loss described above.

Afterwards run the following command to train the DPO model using the provided preference data:

```
$ python run_dpo.py
```

²To understand why we are considering sequences of actions rather than a single action for the next time, recall that 25 actions corresponded to 0.2 seconds of video. If you found it difficult to rank a sequence of 25 actions based on such a short video, imagine ranking the effect of a single action!

³We have prepared a [notebook](#) to illustrate the behavior of `torch.distributions.Independent`.

(d) [2 points (Written)]

Run DPO for 3 different random seeds (you may want to tweak the number of DPO steps to get better results). The plots of the evolution of returns over time are illustrated below.

```
$ python plot.py --dpo-directory results_dpo --output results_dpo/hopper_dpo.png
```

How does it compare to the returns achieved using RLHF? Comment on the pros and cons of each method applied to this specific example.

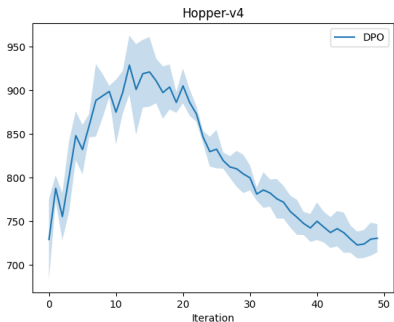
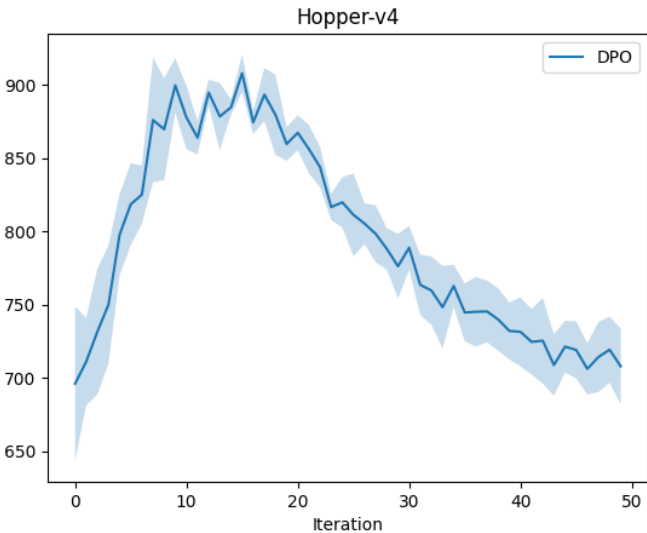


Figure 3: DPO results



The DPO reaches max reward of 950, which better than RLHF but it degraded after it reaches the peak, suggesting overfitting.

Pros and Cons

Method	Pros	Cons
RLHF	More intuitive as it has a two-staged structure.	Unstable. The error in the reward function can be exaggerated by the RL algorithm.
DPO	Avoiding a separate reward function. The RL algorithm can directly optimize the policy.	May overfit the preference data.

(e) [2 points (Written)]

Render a video of an episode generated by the pre-trained policy, and a video of an episode generated by the policy tuned by DPO.

```
$ python render.py --dpo --checkpoint ./results_dpo/[MODEL]/model.pt
```

How do they compare?

To be honest, I don't see significant differences. DPO doesn't seem to improve the performance.

4 Challenges of RLHF

RLHF and DPO leverage humans providing pairwise comparisons.

(a) **[2 points (Written)]**

Who are human raters likely to be that are employed to provide pairwise preferences for generic LLM RLHF/DPO (in terms of socioeconomic profiles)? What ethical issues may this have, and what impact may this have on the resulting trained models?

The raters are often contract worker or from a crowdsourcing platform. These worker typically are from a low-income background. If the raters are not diverse enough, the model may has strong bias on some sensitive topics.

(b) **[2 points (Written)]**

Would RLHF be a good way to train a system to make better medical diagnoses? Why or why not?

No, the medical diagnoses is strongly rely on domain knowledge and must be objective. It shouldn't rely on preferences. Unless the model has acceptable level of correctness, the RLHF may be used to improve the way it structure the response.

(c) **[2 points (Written)]**

There are a number of limitations of RLHF. Please describe one here (different than parts a and b). You may find it interesting to read [“Open Problems and Fundamental Limitations of Reinforcement Learning from Human Feedback”](#) and you are welcome to describe challenges listed there, or make up your own.

RLHF is lack of factuality checks. Human raters may be lack of specific domain knowledge to do factuality checks and prefer answers that is well-written or easier to read, even if they are inaccurate.

This handout includes space for every question that requires a written response. Please feel free to use it to handwrite your solutions (legibly, please). If you choose to typeset your solutions, the —README.md— for this assignment includes instructions to regenerate this handout with your typeset $\text{\LaTeX}$ solutions.

1.a

It's generally very difficult to translate high level objectives into a math mathematical reward function an agent can optimize and it is very difficult to avoid reward hacking that results in an unintended consequences.

Example: Content recommendation agent

The reward function could be number of clicks. The unintended result could be recommending all the clickbait contents instead of high quality, actually interesting contents.

1.b

It's a single legged robot trying to hop forward.

$$\text{reward} = \text{healthy_reward} + \text{forward_reward} - \text{ctrl_cost}$$

Where

healthy_reward : Keep the agent to stand by keeping the torso in a healthy state

forward_reward : Push the agent to move forward by maximize the x-velocity

ctrl_cost : Prevent the agent to aggressively move, i.e. move smoothly by penalize the large control

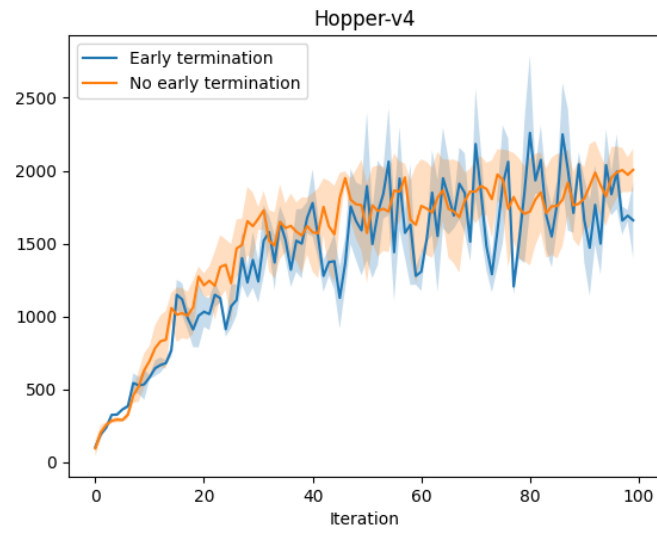
1.c

It means the hopper is standing and not falling over. It is determined by two factors: the torso's height must be above 0.8 meters, and its angle must be within 0.2 radians of vertical.

Advantage of early-termination: speed up training.

Disadvantage of early-termination: can't learn to recover from falling.

1.e



The performance without early-termination is better than with early-termination. It also more consistent, i.e. the standard error is smaller. To get better estimate, we should sample more times by using different random seeds (at least 10-20 times).

1.f

I looked at a rendered video of the agent trained without early-termination.

The agent seem to learn some basic control an able to move forward. But it also keep repeatedly falling and trying to stand up again.

It partially complete the task because it's moving. But it fails to keep the agent in a healthy state. It doesn't seem to be too surprised because the reward function doesn't care about the continuous time it keep in the healthy state.

1.g

I looked at another without-early-termination model.

The agent quickly falling over with very little progress. It still able to crawling forward on the ground but never stand up again. I would consider this a completely failing.

2.a

Let $R_1 = \sum \hat{r}(o^1, a^1) = \sum \phi_w(o^1, a^1)$ and $R_2 = \sum \hat{r}(o^2, a^2) = \sum \phi_w(o^2, a^2)$.

We can express the Bradley-Terry in sigmoid form.

$$\begin{aligned}\hat{P}[\sigma^1 > \sigma^2] &= \frac{e^{R_1}}{e^{R_1} + e^{R_2}} \\ &= \frac{1}{1 + e^{-(R_1 - R_2)}} \\ &= \sigma(R_1 - R_2)\end{aligned}$$

Substitute into loss function

$$L = -(\mu(1) \log(\sigma(R_1 - R_2)) + \mu(2) \log(\sigma(R_2 - R_1)))$$

Let $z = R_1 - R_2$, applying chain rule

$$\begin{aligned}\nabla_w L &= [-\mu(1)(1 - \sigma(z)) + \mu(2)\sigma(z)]\nabla_w z \\ &= [-\mu(1) + \mu(1)\sigma(z) + \mu(2)\sigma(z)]\nabla_w z \\ &= [(\mu(1) + \mu(2))\sigma(z) - \mu(1)]\nabla_w z \\ &= [(\mu(1) + \mu(2))\sigma(R_1 - R_2) - \mu(1)]\nabla_w (R_1 - R_2) \\ &= [(\mu(1) + \mu(2))\sigma(\sum_t \phi_w(o_t^1, a_t^1) - \sum_t \phi_w(o_t^2, a_t^2)) - \mu(1)](\sum_t \nabla_w \phi_w(o_t^1, a_t^1) - \sum_t \nabla_w \phi_w(o_t^2, a_t^2))\end{aligned}$$

2.b

I got the following pairs

3147 (equally preferred): agree, no obvious differences

2.c

Four more pairs:

4191 (equally preferred): agree, no obvious differences

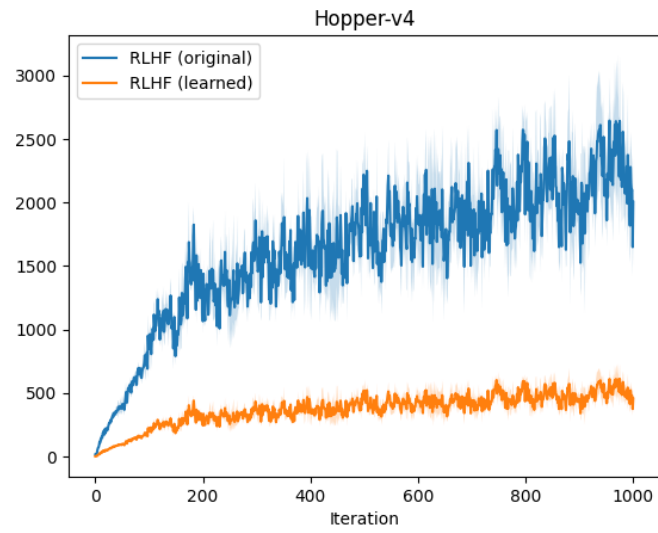
6599 (one is preferred): not agree, the non-preferred one has better ending ankle angle to land

6844 (one is preferred): similar to 6599

7324 (one is preferred): agree, the preferred one landed on its heel, which is a better position to initiate the next hop

I am not confident to say I trust this data because firstly the sample is very short, and I disagree some of the samples.

2.e



They are highly correlated.

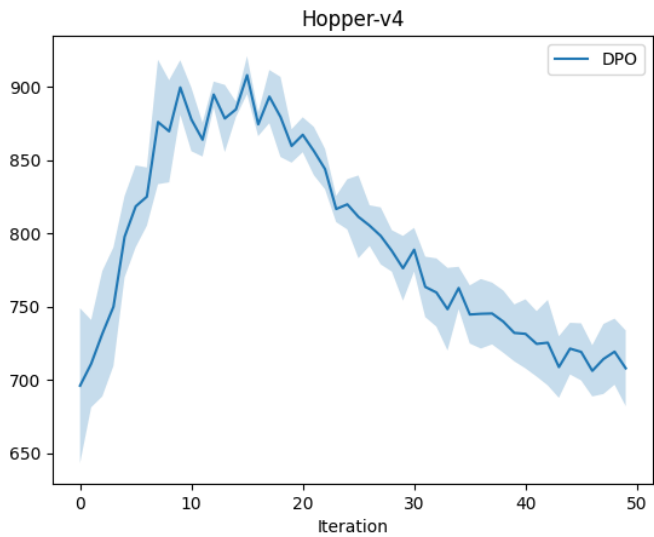
2.f

It does not identifiable because it has very similar trend as the original reward.

2.g

It can perform more hops without falling. I think it is better in terms of hopping but the original policies can move forward more. The RLHF seems working as we prefer a landing position that is easier to initiate next hop.

3.d



The DPO reaches max reward of 950, which better than RLHF but it degraded after it reaches the peak, suggesting overfitting.

Pros and Cons

Method	Pros	Cons
RLHF	More intuitive as it has a two-staged structure.	Unstable. The error in the reward function can be exaggerated by the RL algorithm.
DPO	Avoiding a separate reward function. The RL algorithm can directly optimize the policy.	May overfit the preference data.

3.e

To be honest, I don't see significant differences. DPO doesn't seem to improve the performance.

4.a

The raters are often contract worker or from a crowdsourcing platform. These worker typically are from a low-income background. If the raters are not diverse enough, the model may has strong bias on some sensitive topics.

4.b

No, the medical diagnoses is strongly rely on domain knowledge and must be objective. It shouldn't rely on preferences. Unless the model has acceptable level of correctness, the RLHF may be used to improve the way it structure the response.

4.c

RLHF is lack of factuality checks. Human raters may be lack of specific domain knowledge to do factuality checks and prefer answers that is well-written or easier to read, even if they are inaccurate.